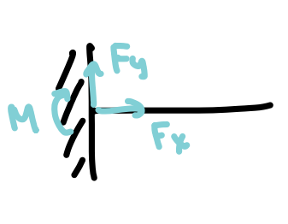
CIV102 Final Exam Review

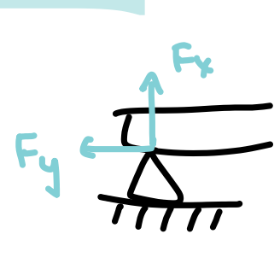
Types of supports

① Fixed end



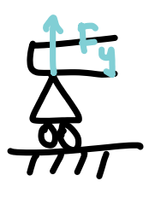
$\Delta x = 0$	$F_x = ?$
$\Delta y = 0$	$F_y = ?$
$\theta = 0$	$M = ?$

② Pin



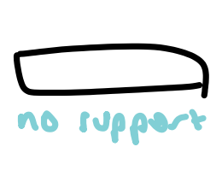
$\Delta x = 0$	$F_x = ?$
$\Delta y = 0$	$F_y = ?$
$\theta \neq 0$	$M = 0$

③ Roller



$\Delta x = ?$	$F_x = 0$
$\Delta y = 0$	$F_y = ?$
$\theta \neq 0$	$M = 0$

④ Free end



$\Delta x \neq 0$	$F_x = ?$
$\Delta y \neq 0$	$F_y = ?$
$\theta \neq 0$	$M = ?$

THINGS TO REMEMBER:

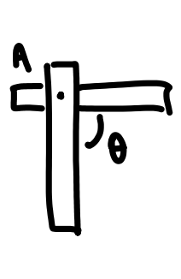
- actual loads in truss :(

  1 MPa = 1 N/mm²

  1 kN/m = 1 N/mm

- distance in MAT to point of interest not point of tangency

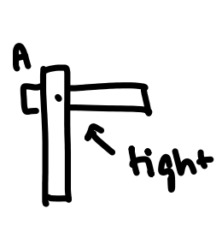
Hinge :



$\theta = ? \Rightarrow M = 0$

$\Sigma M_A = 0$

Rigid / fixed :



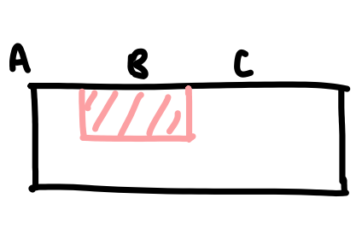
$\theta = \text{constant} \Rightarrow M \neq 0$

$\Sigma M_A \neq 0$

Trusses

assumes all joints are hinges, shear = moment = 0 => only axial loads

Tributary area :



$\text{Load @ B} = \text{Area} \times \underbrace{q}_{\text{pressure (kPa = kN/m}^2)}$

Design of Truss Members

	A	I	$r = \sqrt{\frac{I}{A}}$
Tension	$\geq \frac{2.0F}{\sigma_y}$	no buckling	$\geq \frac{L}{200}$ ← length of longest member
Compression		$\geq \frac{3.0FL^2}{\pi^2 E}$	
	$\left(F \leq \frac{\sigma_y A}{2.0}\right)$	$\left(F \leq \frac{\pi^2 EI}{3.0L^2}\right)$	

HSS in Canada :

$\sigma_y = 350$ MPa

$E = 200000$ MPa

$$\text{Wind pressure} = \frac{1}{2}\underset{\text{air density} \sim 1.2\,\text{kg/m}^3}{\rho}\overset{\text{airspeed}}{v^2}\overset{\text{coefficient of drag}}{C_D}$$

Deflections by Virtual Work

displacement in truss member $= \frac{PL}{EA}$

$$\boxed{\underbrace{F^*}_{\text{virtual}} \cdot \underbrace{\Delta}_{\text{REAL}} = \Sigma \underbrace{P_i^*}_{\text{virtual}} \cdot \underbrace{\Delta_i}_{\text{REAL}} \left(\frac{PL}{EA}\right)}$$

① Solve real Δ in members $\left(\frac{PL}{EA}\right)$

② Apply 1kN @ location of interest, solve forces

③ Apply PVW

Member	P kN	$\varepsilon = \frac{P}{EA} = \frac{\text{kN}}{(\text{MPa})(\text{mm}^2)} \times 10^6 = \text{mm/m}$	L m	$\Delta = \varepsilon L$ mm $\left(\frac{PL}{EA}\right)$	P^* kN (watch sign)	$P^* \cdot \Delta_L$ kN·mm = J
⋮					⋮	⋮
						Total $P^* \cdot \Delta L =$ ___

$$\underbrace{F^*}_{1\text{kN}} \cdot \Delta = \overbrace{\Sigma P^* \cdot \Delta_i}^{J}$$

$\Delta =$ ___ (mm)

Oscillations

Point load @ midspan : $\boxed{f_n = \frac{15.76}{\sqrt{\Delta_0}}}$ ← displacement @ midspan (PVW)

Uniformly distributed load : $\boxed{f_n = \frac{17.76}{\sqrt{\Delta_0}}}$ natural freq. of truss

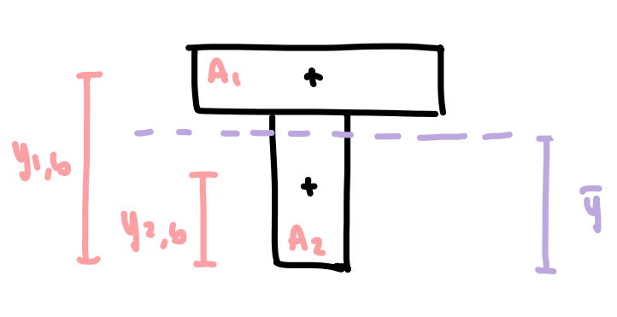
Damping

$$\boxed{DAF = \frac{1}{\sqrt{\left(1-\left(\frac{f}{f_n}\right)^2\right)^2 + \left(2\beta\frac{f}{f_n}\right)^2}}}$$

Beams

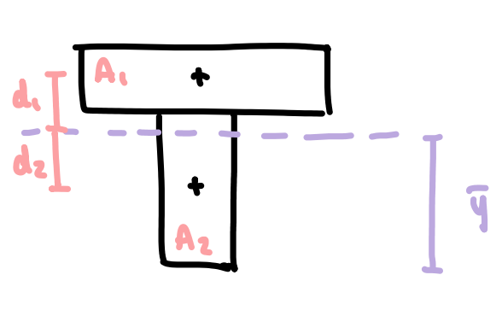
Centroidal axis :

$$\boxed{\bar{y} = \frac{\Sigma A_i \overbrace{y_{i,b}}^{\text{dist. to bottom}}}{\Sigma A_i}}$$



Second moment of area :

$$\boxed{I = \underbrace{\Sigma I_{i,0}}_{\substack{\text{about own centroid} \\ \left(\frac{bh^3}{12}\right)}} + \Sigma A_i \underbrace{d_i^2}_{\text{dist. to } \bar{y}}}$$



$d_1 = |y_{1,b} - \bar{y}|$

$d_2 = |y_{2,b} - \bar{y}|$

$I_{rect} = \frac{bh^3}{12}$

$I_{circle} = \frac{\pi d^4}{64}$ ← diameter

Navier's equation :

$\boxed{\sigma = \frac{My}{I}}$ ← dist. from point to centroidal axis

$\frac{\text{Nmm} \cdot \text{mm}}{\text{mm}^4} = \text{MPa}$

$\frac{\text{kNm} \cdot \text{mm}}{\text{mm}^4} = 10^6\ \text{MPa}$

Curvature :

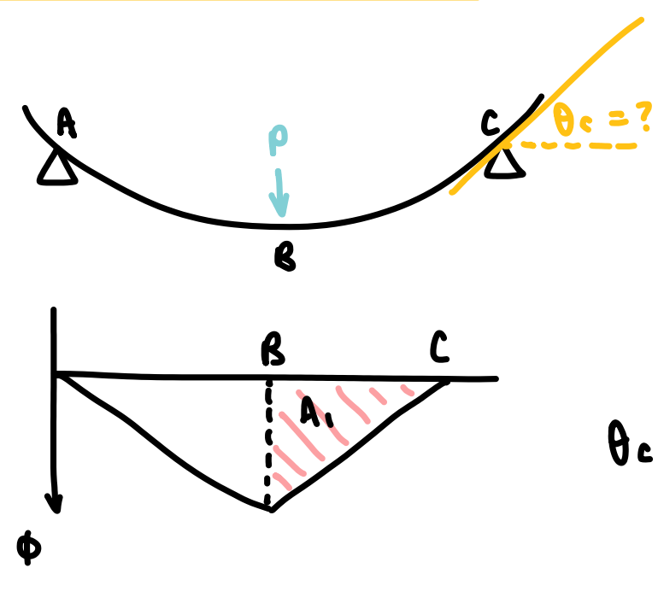
$\boxed{\phi = \frac{M}{EI}}$

$EI = (\text{MPa})(\text{mm}^4) = \text{Nmm}^2$

$\frac{\text{Nmm}}{\text{Nmm}^2} = \text{mm}^{-1}$

MAT #1 :

$\boxed{\theta_B - \theta_A = \int_A^B \phi(x)\,dx}$ change in slope = area under φ diagram



$\theta_c = A_1$ (radians)

MAT #2 :

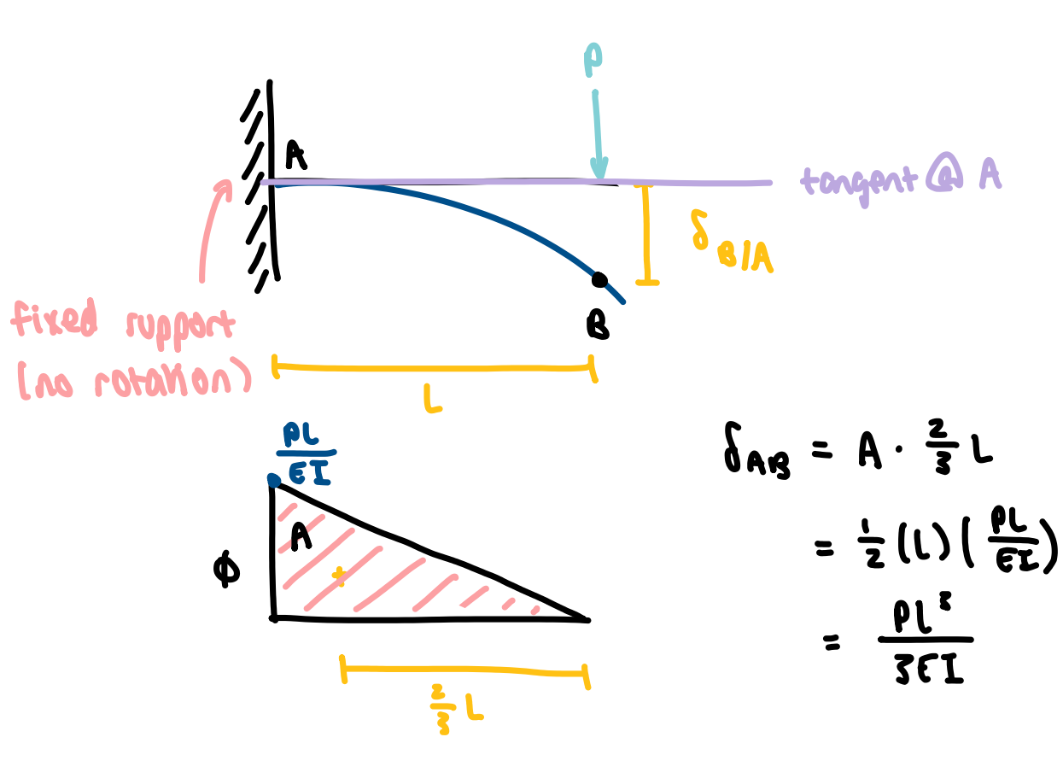
$$\boxed{\delta_{AB} = \int_A^B x\phi(x)\,dx}$$

tangent @ B

$= A_{AB} \cdot \text{dist}$ ← first moment of area of φ diagram

dist: calculated from A

case 1 : known horizontal tangent due to support



$\delta_{AB} = A \cdot \frac{2}{3}L$

$= \frac{1}{2}(L)\left(\frac{PL}{EI}\right)\left(\frac{2}{3}L\right)$

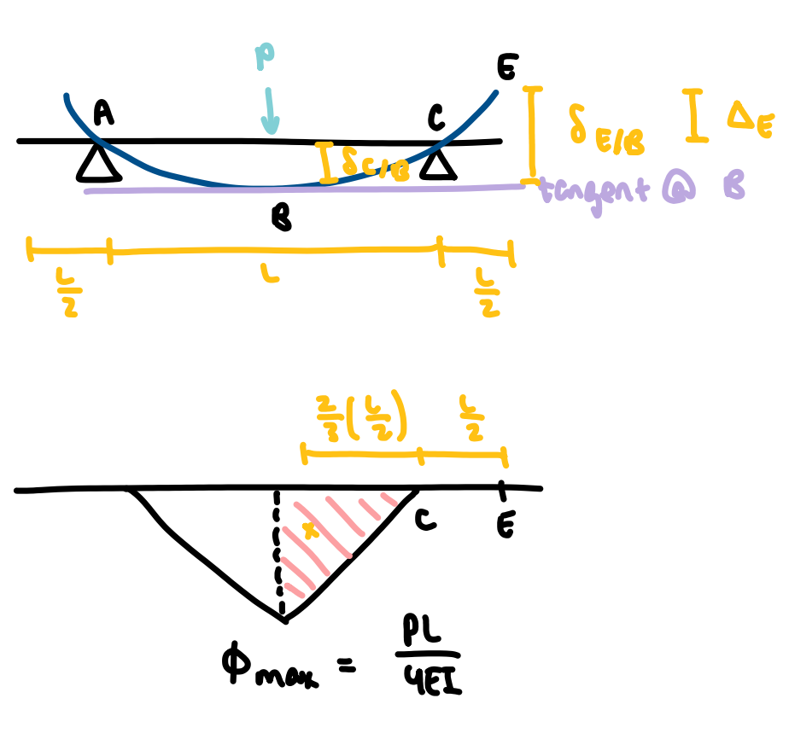
$= \frac{PL^3}{3EI}$

$\theta_{AB} = \theta_B - 0$ ← $\theta_A = 0$

$= \frac{1}{2} \cdot L \cdot \frac{PL}{EI}$

$= \frac{PL^2}{2EI}$

case 2 : known horizontal tangent due to symmetry



$\Delta E = \delta_{E/B} - \delta_{C/B}$

$= \frac{1}{2}\left(\frac{L}{2}\right)\left(\frac{PL}{4EI}\right)\left(\frac{2}{3}\cdot\frac{L}{2}+\frac{L}{2}\right) - \frac{1}{2}\left(\frac{L}{2}\right)\left(\frac{PL}{4EI}\right)\left(\frac{2}{3}\cdot\frac{L}{2}\right)$

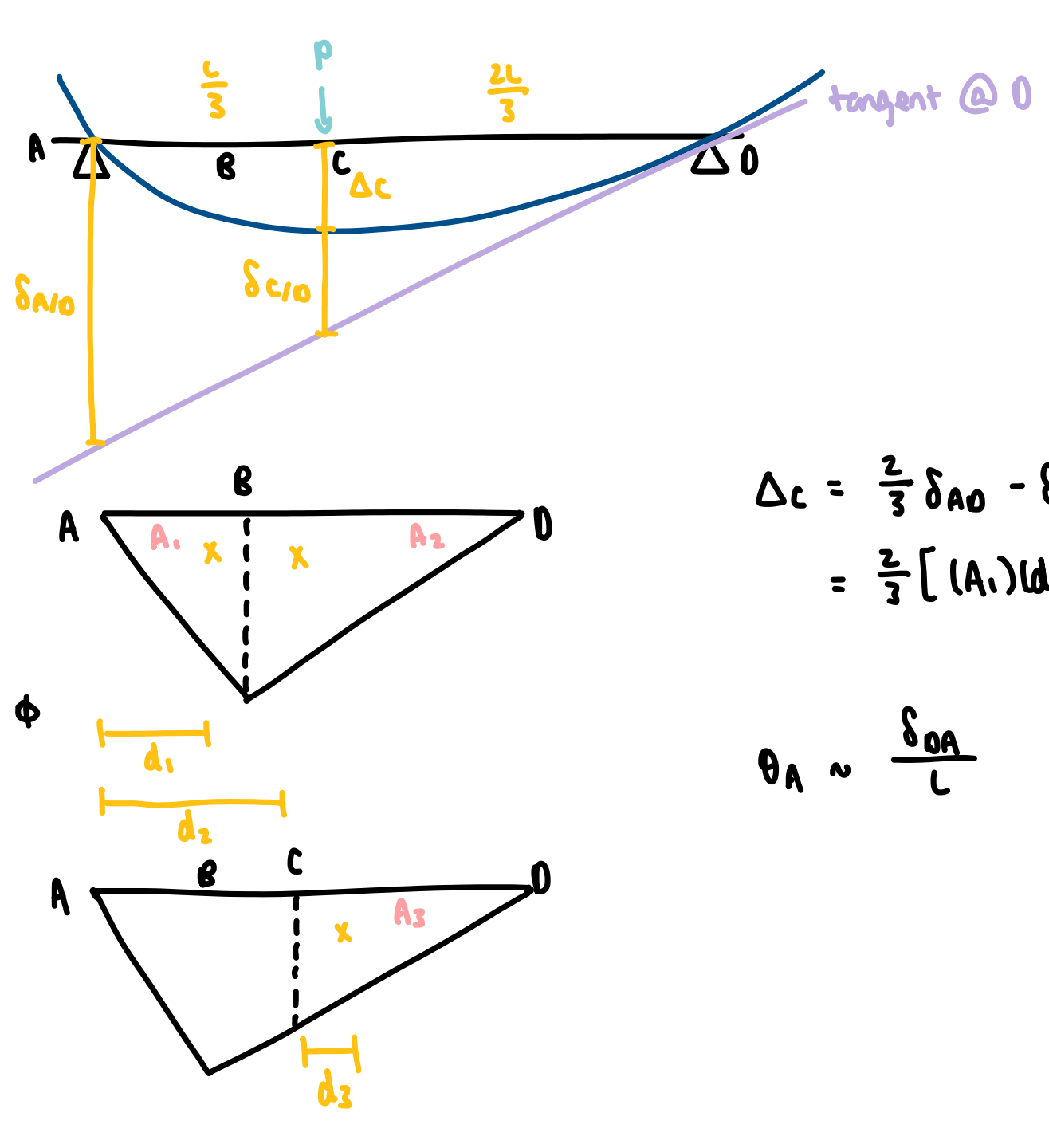
$= \frac{PL^3}{32EI}$

$\theta_{BC} = \theta_C - 0$ ← $\theta_B = 0$

$= \frac{1}{2}\cdot\frac{L}{2}\cdot\frac{PL}{4EI}$

$= \frac{PL^2}{16EI}$

case 3 : no known horizontal tangents



$\Delta_C = \frac{2}{3}\delta_{AD} - \delta_{CD}$

$= \frac{2}{3}[(A_1)(d_1) + (A_2)(d_2)] - (A_3)(d_3)$

$\theta_A \sim \frac{\delta_{DA}}{L}$ from one support to the other support

$$\boxed{\text{for a beam, } F_A = P \cdot \frac{\Delta_A}{\delta_A^*}}$$

F_A: force at support A

Δ_A ← actual displacement @ A

δ_A^* ← displacement with unit load @ A (MVW)

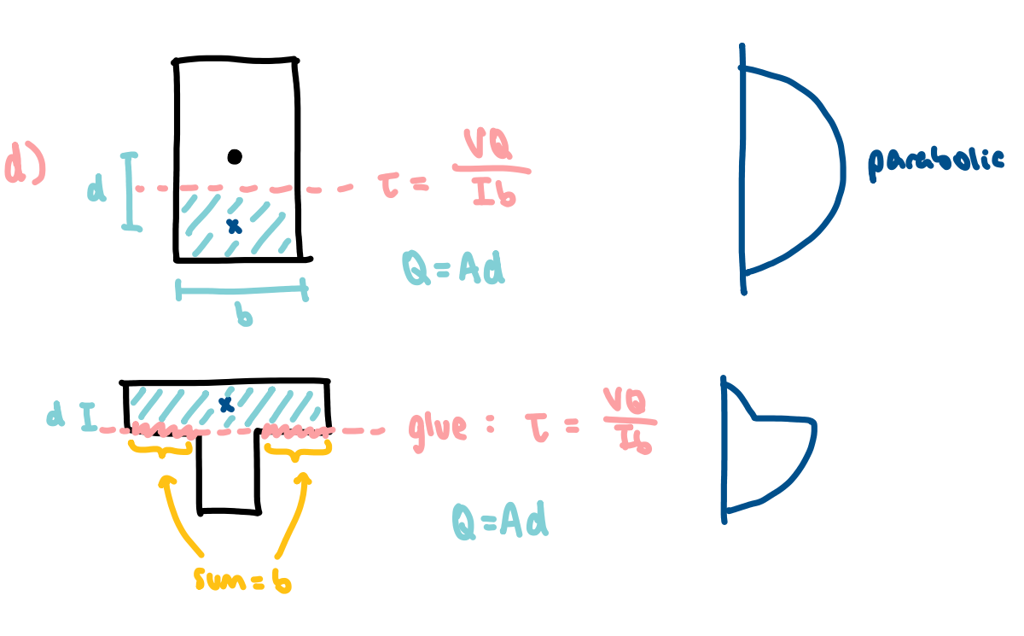
Shear stresses

Jourawski's equation :

$\boxed{\tau = \frac{VQ}{Ib}}$ (max @ centroid)

τ: shear stress

$\boxed{Q = \Sigma A_i d_i}$



Euler buckling :

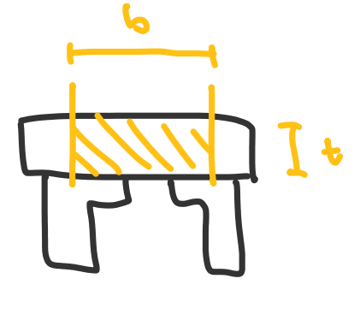
$\boxed{P_{crit} = \frac{\pi^2 EI}{L^2}}$

Local buckling

$\sigma = \frac{My}{I}$

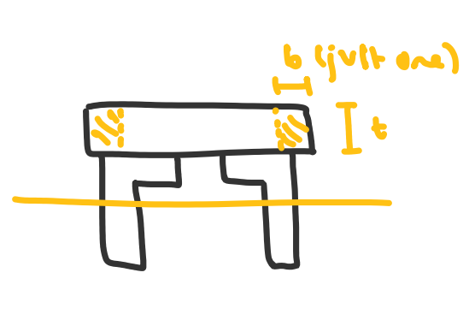
case 1 : restrained on 2 sides (e.g. compressive flange between webs)

$\sigma = \frac{4\pi^2 E}{12(1-\mu^2)}\left(\frac{t}{b}\right)^2$



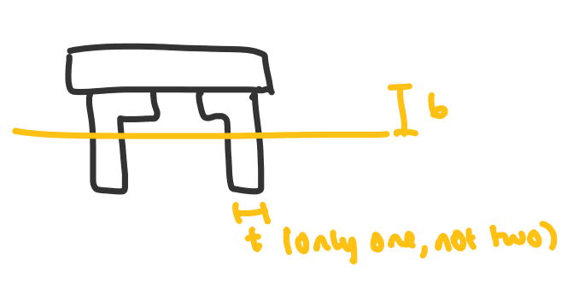
case 2 : restrained on 1 side (e.g. tips of flange)

$\sigma = \frac{0.425\pi^2 E}{12(1-\mu^2)}\left(\frac{t}{b}\right)^2$



case 3 : restrained on 2 sides (e.g. buckling of webs due to flexural stress)

$\sigma = \frac{6\pi^2 E}{12(1-\mu^2)}\left(\frac{t}{b}\right)^2$



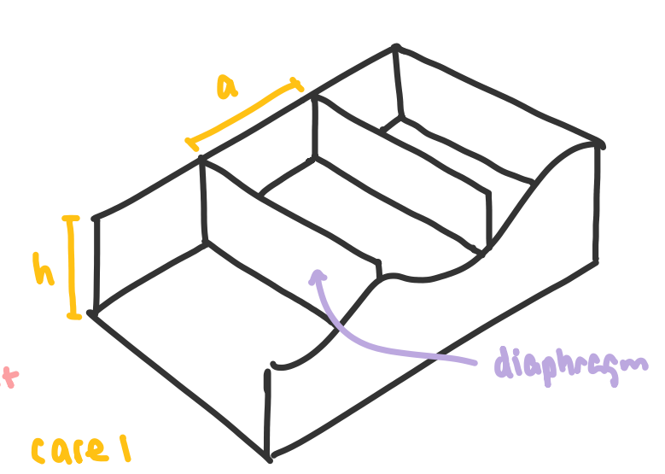
$\tau = \frac{VQ}{Ib}$

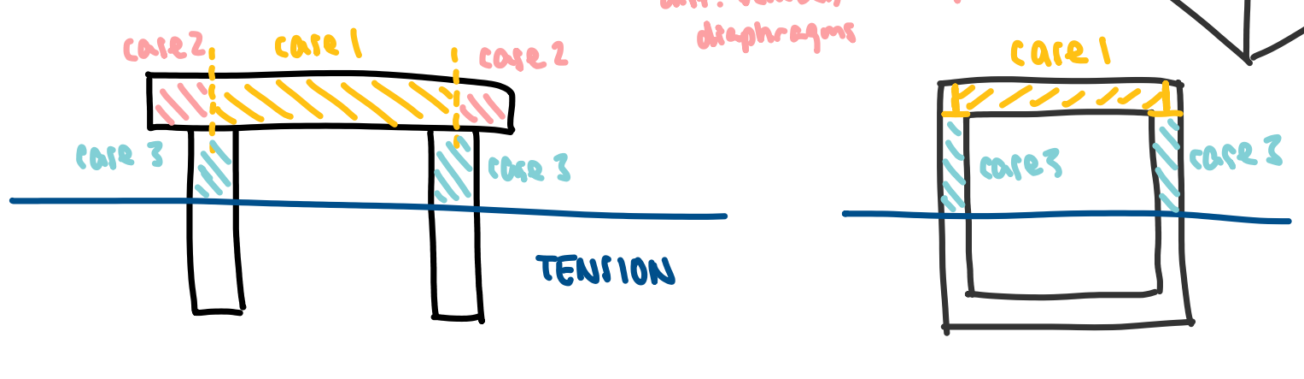
Case 4 : shear buckling of webs

$\tau = \frac{5\pi^2 E}{12\cdot(1-\mu)^2}\left[\left(\frac{t}{a}\right)^2 + \left(\frac{t}{h}\right)^2\right]$

a: dist. between diaphragms

h: height





Concrete

concrete compressive strength (f_c') & concrete tensile strength (f_t')

$f_t' = 0.33\sqrt{f_c'}$

Young's modulus of concrete :

$$\boxed{\underset{\text{MPa}}{E_c} = 4730\sqrt{\underset{\text{MPa}}{f_c'}}}$$

Rebars : $E_s = 200\,000$ MPa, $f_y = 400$ MPa

① Knowns

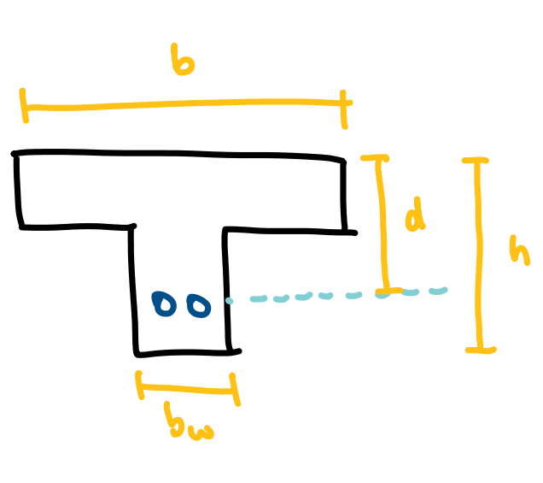
material properties f'_c, f'_y

concrete geometry b_w, d, b

b_w: width of web

d: effective depth, dist. from comp. loc. to centroid of reinforcement

loads → from SFD



$$\boxed{\begin{aligned} n &= \frac{E_s}{E_c} \\ \rho &= \frac{A_s}{b \cdot d} \\ k &= \sqrt{(n\cdot\rho)^2 + 2n\rho} - n\rho \\ j &= 1 - \frac{k}{3} \\ j\cdot d &= \text{lever arm} = d - \frac{kd}{3} \end{aligned}}$$

A_s: area of steel

② Check stress

$A_{s,min} = \frac{M}{\underbrace{0.5f_y}_{\text{allowable (FOS = 2)}}jd}$

Stress in steel : $f_s = \frac{M}{A_s \cdot j \cdot d}$

allowable : $\frac{f_s'}{\text{F.O.S.}} = \frac{f_s'}{2.0}$

Stress in concrete : $f_c = \frac{k}{1-k} \cdot \frac{M}{jd \cdot A_s \cdot n}$

allowable : $\frac{f_c'}{\text{F.O.S.}} = \frac{f_c'}{2.0}$

stress	max stress
f_c	f'_c
f_s	f'_s

③ Check shear

$$\boxed{V_{applied} \leq \frac{V_c}{\text{F.O.S.}} + \frac{V_s}{\text{F.O.S.}} \leq V_{ultimate} = 0.25 f_c' \cdot b_w \cdot j \cdot d}$$

F.O.S.: 2.0

$0.25 f_c' \cdot b_w \cdot j \cdot d$: shear to cause diagonal crushing

members without stirrups : $\left(\frac{A_v f_y}{b_w s} < 0.06\sqrt{f_c'}\right)$

$V_s = 0$

$V_c = \underbrace{\frac{230\sqrt{f_c'}}{1000 + 0.9d}}_{\text{stress}} \underbrace{b_w (j\cdot d)}_{\text{area}}$

in MPa, mm, N

members with stirrups : $\left(\frac{A_v \cdot F_y}{b_w \cdot s} \geq 0.06\sqrt{f_c'}\right)$

$V_c = 0.18\sqrt{f_c'}\, b_w \cdot j \cdot d$

$V_s = \frac{A_v \cdot f_y \cdot (j\cdot d)}{s}\cot\theta$

divide F.O.S. = 2.0

θ = 35° in this class

A_v = total cross sectional area of vertical bars (# bars × A_{bar})

Choosing value of s :

a) $\frac{A_v \cdot F_y}{b_w \cdot s} = 0.06\sqrt{f_c'}$

b) $s \leq 0.63d$

c) $s \leq 600$ mm

take the smallest